RICE GRAIN VARIETY CLASSIFICATION

1. Problem and Objective

Rice varieties have different shapes, sizes, and appearances, making manual identification time-consuming. This project uses a Convolutional Neural Network (CNN) to classify rice grain images automatically. The objective is to identify five rice varieties: Arborio, Basmati, Ipsala, Jasmine, and Karacadag. The model learns visual features from images and predicts the correct rice variety. This can help improve the speed and consistency of rice classification.

2. Dataset and Pre-processing

The Rice Image Dataset was obtained from Kaggle and contains five rice varieties: Arborio, Basmati, Ipsala, Jasmine, and Karacadag. A balanced subset of 500 images per class was selected, giving a total of 2,500 images. The images were resized to 128 × 128 pixels and pixel values were normalized between 0 and 1. The dataset was divided into 70% training, 15% validation, and 15% testing data. Data augmentation, including rotation, zooming, and horizontal flipping, was applied to training images to improve generalization.

3. Methodology:

A CNN model was developed to learn important visual features from rice grain images. The model contains convolutional layers, max-pooling layers, a flatten layer, dense layers, and a dropout layer. The final softmax layer predicts one of the five rice varieties. The Adam optimizer and categorical cross-entropy loss function were used for training. Early stopping was applied to help reduce overfitting.

Parameter	**Configuration**
Model	Convolutional Neural Network (CNN)
Image size	128 × 128
Images per class	500

4. Results:

Evaluation Set	Accuracy	Weighted F1-score	Interpretation
Training set	92.20%	0.9920	Shows how well the model learned from training images.
Validation split	98.40%	0.9840	Measures performance on unseen validation images during training.
Test set	98.00%	0.9800	Evaluates the model's performance on unseen rice grain images.

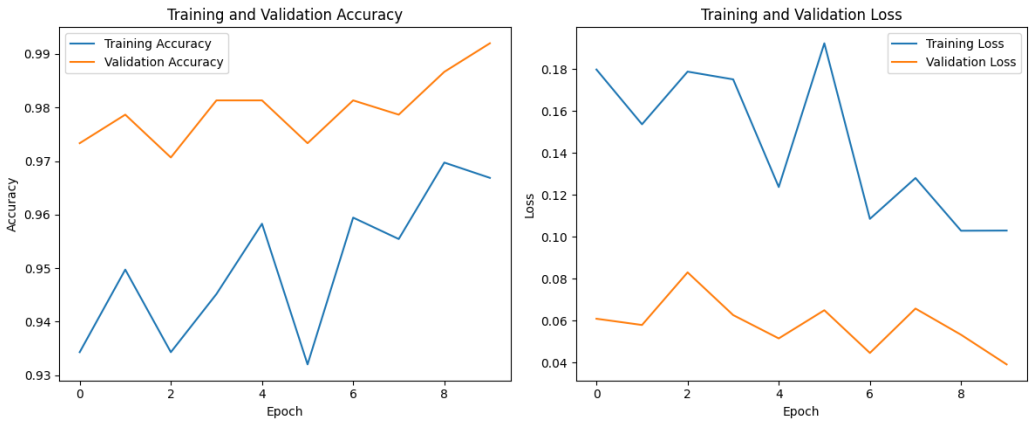


Figure 1: Training and validation accuracy and loss

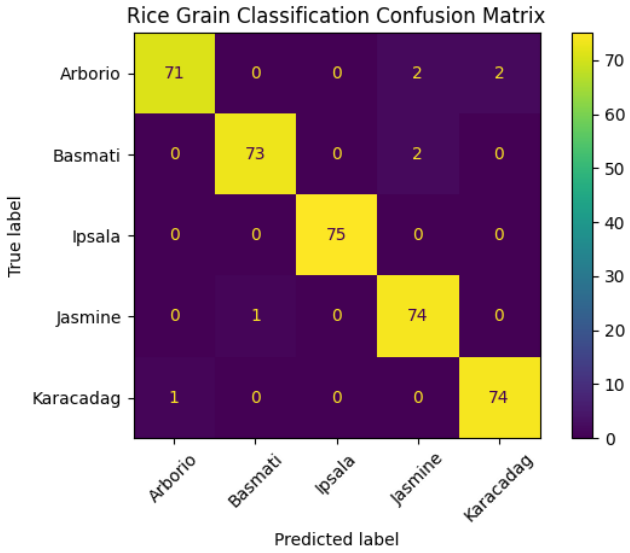


Figure 2: Rice Grain Classification Confusion Matrix

5. Analysis and Discussion:

The training and validation graphs help show how effectively the CNN learns from the images. Test accuracy indicates how well the model classifies unseen rice grain images. The confusion matrix shows which varieties are classified correctly and which are confused with one another. Misclassification may occur because some rice varieties have similar shapes, colours, or sizes. The actual evaluation metrics are needed to determine the model's performance and whether overfitting occurred.

6. Real-world Relevance, Limitations and Ethics

Automatic rice classification can support rice quality inspection, agricultural processing, packaging, and grain sorting. It may reduce manual effort and help provide more consistent classification. However, the model was trained on a limited subset of images and may not perform equally well under different lighting or camera conditions. Similar-looking varieties

may also be misclassified. The model should be tested on real-world images before practical use. Dataset attribution should be maintained, and predictions should be checked before making important quality decisions.

7. Conclusion and Future Work

This project demonstrates the use of a CNN to classify five rice grain varieties from images. The model's performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. In future, the model can be improved by training on more diverse images, tuning hyperparameters, and testing it on real-world rice samples.

8. Result

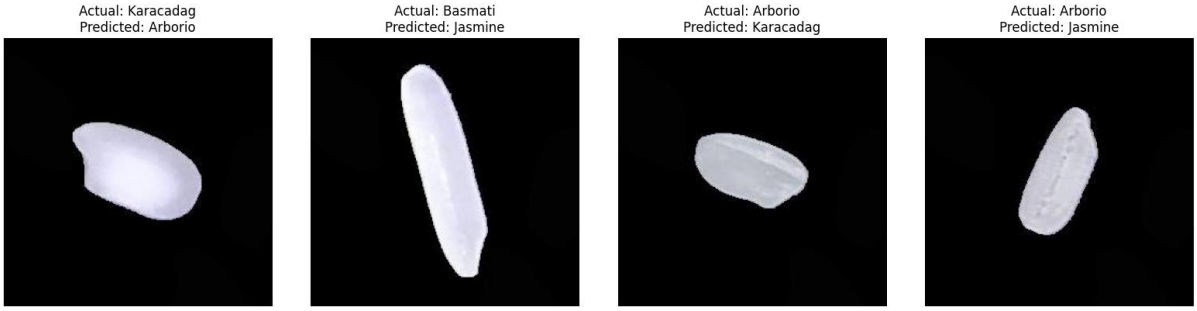


Figure 1: Sample Rice Grain variety classification

Classification Report:

	precision	recall	f1-score	support
Arborio	0.99	0.95	0.97	75
Basmati	0.99	0.97	0.98	75
Ipsala	1.00	1.00	1.00	75
Jasmine	0.95	0.99	0.97	75
Karacadag	0.97	0.99	0.98	75
accuracy			0.98	375
macro avg	0.98	0.98	0.98	375
weighted avg	0.98	0.98	0.98	375

Figure 2: Classification report

9. Dataset Citation

Koklu, M. *Rice Image Dataset*. Kaggle. Available at:

https://www.kaggle.com/datasets/muratkokludataset/rice-image-dataset